Recall First, Then Let the Model Choose: Gemma 4 as a Bug-Localization Selector, Tested on Post-Cutoff Issues

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Kaggle · Google Gemma 4 Developer Agent Paper Track · November 2026

Abstract

Before a coding agent can fix a bug it must find the file to edit. We compare two ways of spending Gemma-4-12B-it on this step. As a **scorer** — ranking code by how much the issue lowers the model's cross-entropy on it (NAMI) — the model is weaker than BM25. As a **selector** — shown the issue and a shuffled list of candidate paths, asked to name the likeliest files — it is strong: on the *same* BM25 candidates on 57 SWE-bench Lite instances it places the gold file in the top-5 on 36 instances against NAMI's 15, and on a BM25 ∪ dense pool on 84%, against 65% for dense retrieval and 75% for an oracle choosing the better retriever per instance. Because SWE-bench predates Gemma 4, we then test what the model brings. Removing or swapping the issue drops the model's own top-5 picks to 21% and 14%, and ranking the same candidates by their paths alone reaches 54%; on 120 SWE-bench-Live issues created after the model's cutoff, with the pipeline frozen, the selector beats dense retrieval at top-1 (61% vs 38%, significant) and leads at top-5 (85% vs 72.5%, not significant after correction). One limit remains: on new issues that never name the gold file it shows no detectable advantage over dense retrieval.

1. Introduction

A file that never reaches an agent's context window cannot be fixed, and most repository-level agents localize in two stages: retrieve candidates, then narrow them down. We began with an information-theoretic bet about the second stage: an issue should lower a model's cross-entropy on exactly the faulty code, giving a training-free relevance score. The bet lost; its controls led to three findings:

1. **On identical candidates, choosing beats scoring.** One listwise call puts the gold file in the top-5 on 36/57 instances where likelihood scoring of the same 25 files manages 15 (§4.1).
2. **Pool recall is the next bottleneck.** BM25's top-25 contains the gold file on 63% of instances; adding a 90 MB embedder's top-25 raises this to 89%, and the selector converts that into 84% top-5 (§4.2–4.3).
3. **The selector's skill survives a post-cutoff test.** Controls and 120 issues created after its training cutoff separate reading from recall and show where the gain is not (§4.4).

2. Related work

Likelihood for retrieval and defects. Open LLMs rank documents by query likelihood [1]; null-prompt calibration [2] and context–question PMI [3] are related; surprisal as a defect signal goes back to [4, 5]. **LLM rerankers and selectors.** Listwise reranking [6] has been adapted to issue localization by LLM-BL [7], CodeRankLLM [8] and SweRank [9], the latter two fine-tuned; Agentless [10] narrows hierarchically and LocAgent [11] searches a code graph. Our selector is a zero-shot member of this family; we add a same-pool comparison with likelihood scoring, a pool-ceiling decomposition and a memorization audit. **Memorization.** Models identify SWE-bench file paths from the issue alone far more often than on other repositories

[12, 13]. Loc-Bench [11] starts in October 2024, inside Gemma 4's training window (January 2025 cutoff [16]), so we use SWE-bench-Live [17] issues created from April 2025.

3. Methods

Likelihood scoring (NAMI). For issue I and an AST chunk $C = (t_1..t_N)$, a frozen causal LM gives $H(C \mid X) = -\frac{1}{N}\sum_i \log_2 P(t_i \mid X, t_{<i})$. NAMI $= H(C \mid I_{null}) - H(C \mid I)$ — a pointwise, PMI-style contrast — with I_{null} a bug report from another, non-scientific repository matched to ±30% of I's length. Files take their best chunk's score. Controls: query likelihood (QLM), mean $\log_2 P(I \mid C)$; a no-null ablation ranking by $-H(C \mid I)$; reciprocal-rank fusion (RRF, k=60) with BM25. Applying the LM head to 128 positions at a time keeps memory flat on one A100 (bf16).

Retrievers. Tree-sitter chunks (functions and classes; whole files ≤4,000 characters when nothing parses) over all Python files outside `tests/` and `docs/` at the base commit. BM25 (`rank_bm25` Okapi defaults, lower-cased whitespace tokens, chunks ≤6,000 characters) and `all-MiniLM-L6-v2` [15] cosine similarity (default 256-token window), each max-aggregated to files.

Listwise selection. Pool = BM25 top-25 ∪ dense top-25 (28–49 files, mean 41.8), shuffled with a fixed per-instance seed so the model cannot copy retrieval order. `google/gemma-4-12b-it` (bf16) receives the repository name, the issue (first 4,000 characters; hints are not used) and the numbered list — paths only, or paths plus up to 15 class/function names per file — through its default chat template as one user turn (no system prompt, no thinking flag), greedy, ≤80 new tokens (Appendix A). Its first 10 distinct valid numbers form the ranking (all 114 Lite and 119/120 post-cutoff answers gave 10); unnamed pool files follow in BM25 order. A call costs about 1.1k prompt tokens and 4.6 s on an A100 (paths only) or 3.8k tokens and 7.6 s (with names).

Data and statistics. 61 SWE-bench Lite [14] instances drawn round-robin under fixed per-repository quotas (django 20, others ≤8); 57 completed likelihood scoring (4 ran out of memory and are excluded from every comparison): django 19, sympy 8, astropy 6, requests 6, matplotlib 5, pylint 5, scikit-learn 5, flask 3. Each has one gold file. An issue *names the file* if the gold file's base name without `.py` (e.g. `autoreload`) occurs in the issue text, case-insensitively, ignoring the stems `__init__`, `base`, `utils`, `core` and stems of ≤2 characters. We report hits on identical instances with uncorrected 95% bootstrap CIs (10,000 resamples) and exact two-sided McNemar tests on paired hits (+b/−c = instances only the first/second method gets right). Tests are Holm-corrected within seven section-level families (§4.1; in-pool rerankers; §4.3; controls; probe; path-only; post-cutoff — 54 tests, all in the released statistics output); every result we call significant survives that correction, and we mark those that do not. *Protocol change:* the planned pool paired BM25 with CodeRankEmbed, which failed at its first encoding; the likelihood methods therefore rerank BM25's top-25 on every instance.

4. Results

4.1 On the same candidates, choosing beats scoring

ranking BM25's top-25 files (57 instances)	top-1	top-5	top-10
NAMI (per-token)	5%	26%	40%
NAMI total-bits / QLM / no-null ablation	11 / 9 / 5%	23 / 25 / 26%	47 / 46 / 42%
Random order of the 25 (expected)	3%	13%	25%
BM25's own order	28%	51%	56%
Gemma selects from the same 25	**47%**	**63%** [51,75]	63%

Gemma beats NAMI at top-5 (+21/−0) and top-1 (+24/−0, both $p<0.0001$) and BM25's order (top-1 +12/−1, $p=0.003$; top-5 +7/−0, $p=0.016$). On the 36 instances whose gold file is in this pool, Gemma's top-5 contains it on all 36; BM25's order on 29 (81%), NAMI's on 15 (42%). Likelihood scoring is therefore well above random but weaker than BM25 (in-pool top-5 +3/−17, $p=0.003$) and adds nothing: the no-null ablation also reaches 15/57 (on different instances, +6/−6), and fusing NAMI into BM25 lowers BM25's top-5 from 29 to 21 hits (+0/−8, $p=0.008$).

4.2 Pool recall is the next bottleneck

On 21/57 instances the gold file is not in BM25's top-25 at all, and dense retrieval ranks 15 of those in its own top-25. **BM25-25 ∪ dense-25 contains the gold file on 51/57 instances (89%) instead of 36 (63%)** (Figure 1a). NAMI was not re-scored on the union pool (about 10 A100-hours), but chunks are scored independently of the other candidates, so a larger pool cannot raise the gold file's rank: NAMI's union top-5 is at most 15 + 15 = 30/57 (53%).

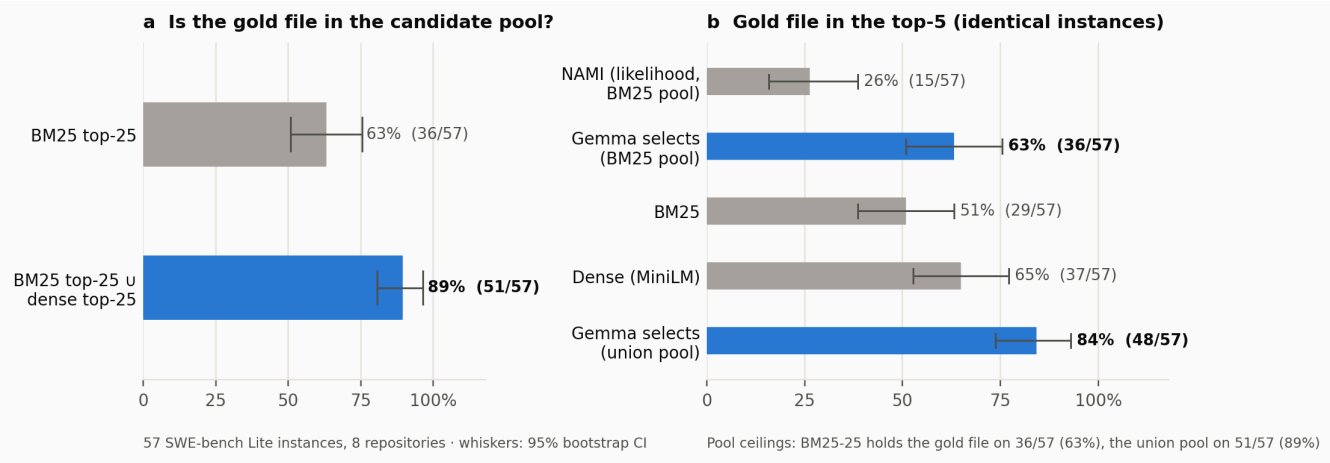


Figure 1. (a) Instances whose gold file is in the candidate pool. (b) Gold file in the top-5 on the same 57 instances. Whiskers: 95% bootstrap CIs.

4.3 A high-recall pool and a selector

union pool (identical 57 instances)	top-1	top-5	top-10	MRR
BM25	28%	51%	56%	0.38
Dense	44%	65%	79%	0.54
RRF(BM25, dense)	35%	58%	74%	0.46
Oracle: better of BM25/dense per instance	54%	75%	84%	—
Gemma, paths only	61% [49,74]	**84%** [74,93]	88%	0.72
Gemma, paths + names	**68%** [56,81]	**84%** [74,93]	88%	0.76

Gemma's top-5 beats dense (paths +13/−2, p=0.007; with names +12/−1, p=0.003) and BM25 (+20/−1, p<0.0001). At top-1 the gain is significant with names (+19/−5, p=0.007) but not with paths only (+18/−8, p=0.08); the two Gemma conditions do not differ (+5/−1, p=0.22). The pool already contains dense's top-25, so the gap is choice, not recall: of the 51 gold files the pool holds, Gemma's top-5 recovers 94%, dense's 73%. Two further shuffles of every candidate list give top-5 of 48 and 49; 53/57 instances have the same top-5 outcome in all three orders. A 4.6 s call replaces a median 205 s of NAMI scoring per instance (323 s with QLM).

4.4 What does the model bring?

SWE-bench predates Gemma 4, so the selector might recall answers.

Without the issue the selector collapses. With the same pools and shuffles, replacing the issue by a placeholder puts the gold file among the model's top-5 picks on 12/57 (21%), and another issue from the same repository on 8/57 (14%), against 48/57 with the real issue (+36/−0, +40/−0). We count only the model's own picks: without an issue it often names fewer than ten files, and the usual fill-in order derives from the real issue. **Paths alone are not enough.** Ranking the same pool by BM25 or MiniLM over the path strings — what the selector sees — gives top-5 31/57 and 24/57 against Gemma's 48 (+18/−1, +24/−0); even a per-instance oracle choosing between dense and path-only BM25 only matches the selector (45/57; post-cutoff 105/120 vs 102).

Issue-only, hard options. On all 57 instances we give the model only the issue and ten paths — the gold file and the nine highest-ranked non-gold files of BM25 and dense — and score other readers on the identical options (Figure 2). Gemma picks the gold file on 44 (77%), against 25 for dense and 16 for BM25 over the files' code (their ordinary top-1 counts), and 18 for matching path tokens against the issue text with ties broken by BM25 order (+27/−1). On the 23 issues that do not name the gold file, Gemma is right on 16, path matching on 1 (+15/−0, p=0.0001) and dense on 8 (+11/−3, p=0.057). On Lite, lexical name matching does not explain the selector.

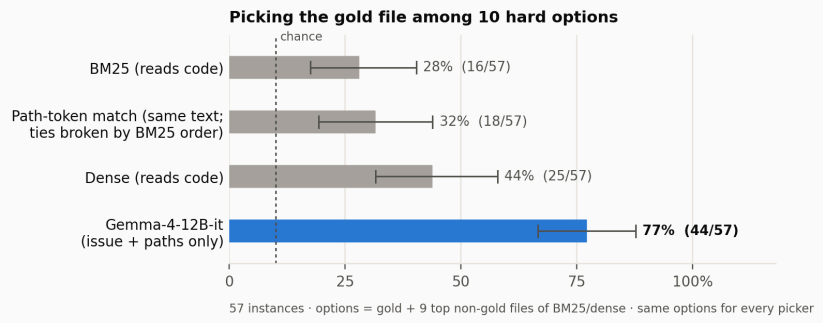


Figure 2. Issue-only 10-way choice with hard distractors on the 57 instances. Distractors are the top non-gold files of BM25 and dense, so those readers score their ordinary top-1; only Gemma is told the answer is among the options. Chance: 10%.

After the cutoff. We sampled 120 SWE-bench-Live issues created between April and August 2025 (at most 3 per repository, 84 repositories), three months past Gemma 4's January 2025 pre-training cutoff (later post-training data is undisclosed), and ran the frozen pipeline: same retrievers, pool recipe, seed rule and prompt, no re-tuning, nothing executed. Gold files are the non-test Python files changed by the fix (47 instances change several; a hit is any gold file). These repositories are smaller (median 127 Python files against 718), so the union pool holds a gold file on 118/120.

post-cutoff (120 instances)	top-1	top-5	top-10
BM25	33%	55%	69%
Dense	38%	72.5%	79%
Gemma selects	**61%** [53,69]	**85%** [78,91]	**95%**

The issue again does the work: Gemma's 102 top-5 hits compare with 37 for its own picks without the issue (+67/−2), 31 with another issue (+74/−3; same repository for 63 instances, otherwise the next sampled one) and 72 for path-only BM25 (+33/−3). Gemma beats dense at top-1 (+41/−14, p=0.0004), also on the 73 single-file instances (42 vs 23, p=0.003); at top-5 the gain (+27/−12, p=0.024; single-file 60 vs 50, p=0.09) does not survive correction. Its top-1 gain comes entirely from issues that name a file (54 vs 27 of 71). Its top-5 advantage over dense is +12.5 points [+2.6, +21.8] (95% CI, resampling whole repositories), against +19.3 [+9.7, +34.1] on Lite. The top-1 gain shows no sign of fading with distance from the cutoff (+24 points in April–May, +22 from June). Repository size may explain part of the smaller top-5 gain: it is +8 points on the 60 instances from smaller repositories and +17 on the 60 from larger ones (n.s.). The limit is the name-free slice:

top-5 gain over dense	issue names a gold file	issue does not
Lite (before cutoff)	+21 points (n=34)	+17 (n=23)
Post-cutoff	+23 (n=71)	**−2** [−18, +14] (n=49)

On new name-free issues Gemma shows no detectable advantage over dense (top-1 19 vs 19), though it still beats path-only BM25 (top-5 36 vs 17, +20/−1). The drop from +17 is what memorization of Lite would produce, though at these sizes the difference is not significant (p>0.1).

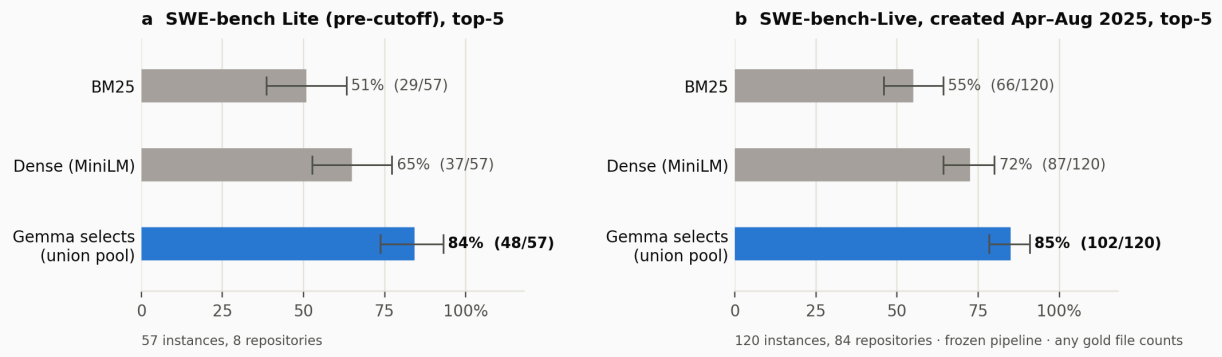


Figure 3. Top-5 before and after Gemma 4's training cutoff, same frozen pipeline. Whiskers: 95% bootstrap CIs.

> **Lessons for agent builders.** (1) Spend the LLM on choosing, not scoring: one 1k-token call over a shuffled path list beat 3–5 minutes of likelihood scoring on the same files. (2) Buy recall before ranking: a 90 MB embedder's top-25 lifted the pool ceiling from 63% to 89%. (3) Shuffle candidates with a fixed seed and report the pool ceiling. (4) Before trusting a public-benchmark number, run an empty-issue control, a path-only baseline (minutes), a post-cutoff slice (an hour) and a name-free slice — gains concentrate where the issue names the file. (5) Gate evaluation harnesses on the official fix: ours caught silent environment failures on the first instance.

5. Reproducibility and disclosure

Code, data and a CPU notebook regenerating every table and test (tag v1.0):

github.com/HussamHaroon/RecallFirst

Released: all per-instance results and option lists, every script, the patch harness and an engineering log. Seeds: candidate shuffle `"<instance>|listwise"`; bootstrap and post-cutoff sample 20261107 (SWE-bench-Live as of 7 November 2026). On 24 earlier-study instances, re-running the pre-fix scorer on another VM 13 days later reproduced every likelihood gold rank. We disclose the padding bug, the CodeRankEmbed fallback and two rounds of harness bugs.

6. Limitations

One model, one embedder, Python only; 57 Lite instances in 8 repositories, too few for per-repository claims. Post-cutoff repositories are smaller, so compare methods within a set, not rates across sets. The name-free conclusions rest on 23 and 49 instances. Path-only baselines are far weaker than the selector, but we did not score paths pointwise with the LM. The scorer used the instruction-tuned checkpoint and max-over-chunks aggregation, neither ablated; baselines are untuned BM25 and a small embedder.

7. Conclusion

Asking Gemma-4-12B-it how surprised it is by code localizes bugs worse than BM25. Asking it to choose among good candidates does — on the same candidates, on a higher-recall pool, and on issues written after it was trained. Its edge over dense retrieval is largest when the issue points at a file; on new issues that do not, it shows no detectable edge — the open problem.

Acknowledgments

Code, experiments and text were produced with substantial AI assistance (coding agents) under the author's direction; all results reproduce from the released repository.

Appendix A. Selector prompt (verbatim)

Bracketed parts appear only with names. The probe uses the same header, a lettered list and “Which single file is most likely modified to fix this bug? Answer with just the letter.”

```
Below is a bug report from the {repo} project, followed by a numbered list of {n}
candidate source files from the repository [(with the classes/functions each file
defines)]. Rank the 10 files most likely to be modified to fix this bug, most likely
first. Answer with only the 10 numbers, comma-separated.

--- BUG REPORT ---
{issue}

--- CANDIDATE FILES ---
1. django/db/models/query.py  [QuerySet, QuerySet.filter, ...]
2. django/utils/autoreload.py  [...]
...

Answer (10 numbers, most likely first):
```

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